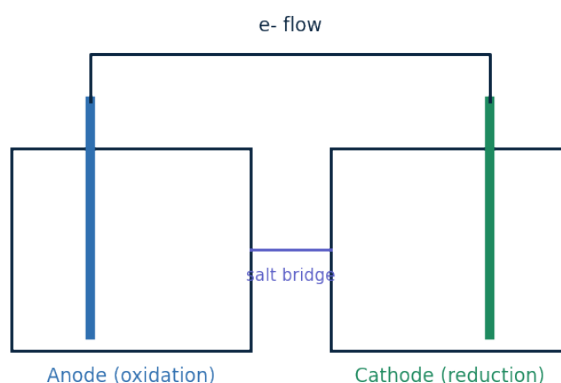


Electrochemistry

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- A galvanic (voltaic) cell converts chemical energy into electrical energy via a spontaneous redox reaction; oxidation occurs at the anode (negative terminal), reduction at the cathode (positive terminal).
- Standard electrode potential (E° -standard) measures a half-cell's tendency to be reduced, measured relative to the standard hydrogen electrode (SHE, $E=0$).
- Cell potential: $E(\text{cell}) = E(\text{cathode}) - E(\text{anode})$ (using standard reduction potentials). A positive $E(\text{cell})$ indicates a spontaneous reaction.
- Nernst equation relates cell potential to concentration: $E = E^\circ - (RT/nF) \ln(Q)$, or at 25 degrees C, $E = E^\circ - (0.0591/n) \log(Q)$.
- Faraday's laws of electrolysis: mass deposited is proportional to the quantity of charge passed (1st law) and to the equivalent weight of the substance (2nd law).
- Conductivity (κ) measures a solution's ability to conduct electricity; molar conductivity (Λ_m) = $\kappa \times 1000 / \text{molarity}$, and increases with dilution.
- Kohlrausch's law: at infinite dilution, molar conductivity of an electrolyte is the sum of the individual ionic conductivities of its cation and anion.



A galvanic cell: oxidation at the anode, reduction at the cathode, connected by a salt bridge.

2. Formula Sheet

Cell potential	$E(\text{cell}) = E(\text{cathode}) - E(\text{anode})$
Nernst equation (25 C)	$E = E\text{-standard} - (0.0591/n) \log(Q)$
Faraday's first law	mass deposited = $Z \times I \times t$ ($Z =$ electrochemical equivalent)
Faraday constant	$F = 96500 \text{ C/mol}$
Relation of Delta G to E(cell)	$\Delta G = -nFE(\text{cell})$
Molar conductivity	$\Lambda_m = \kappa \times 1000 / M$
Kohlrausch's law	$\Lambda_m(\text{infinite}) = \lambda_{+}(\text{cation}) + \lambda_{-}(\text{anion})$
Equivalents deposited	moles of electrons = It / F

3. Practice Problems (22 questions)

1. A cell has $E(\text{cathode}) = +0.34 \text{ V}$ and $E(\text{anode}) = -0.76 \text{ V}$. Find the cell potential.
2. State whether a reaction with $E(\text{cell}) = +1.10 \text{ V}$ is spontaneous, and explain why.
3. Find the number of moles of electrons required to deposit 1 mole of Cu from CuSO_4 solution ($\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$).
4. A current of 2 A is passed through a solution for 1 hour. Find the total charge passed (in coulombs).
5. State Faraday's first law of electrolysis and give its mathematical expression.
6. Find ΔG for a cell reaction with $E(\text{cell}) = 1.1 \text{ V}$ and $n = 2$ ($F = 96500 \text{ C/mol}$).
7. Explain the difference between a galvanic cell and an electrolytic cell.
8. Using the Nernst equation, explain how cell potential changes as the concentration of the reactant ions decreases.
9. Find the equivalent weight of Cu deposited in electrolysis (atomic mass of Cu = 63.5, $n=2$).
10. State Kohlrausch's law and explain its use in finding the molar conductivity of a weak electrolyte at infinite dilution.
11. A standard hydrogen electrode has $E = 0 \text{ V}$ by definition. Explain why it is used as the reference electrode.
12. Find the amount of charge required to deposit 63.5 g (1 mole) of copper from CuSO_4 solution ($n=2$, $F=96500 \text{ C/mol}$).
13. Explain why molar conductivity increases with dilution for both strong and weak electrolytes.
14. State the relationship between Gibbs free energy change and cell EMF, and explain its significance for spontaneity.
15. Find the cell potential for a cell with $E(\text{cathode}) = -0.44 \text{ V}$ and $E(\text{anode}) = -0.76 \text{ V}$ (both standard reduction potentials).

16. Explain what is meant by a 'salt bridge' in a galvanic cell and state its function.
17. A current of 5 A flows for 2 hours through an electrolytic cell. Find the number of moles of electrons passed ($F = 96500 \text{ C/mol}$).
18. State the difference between specific conductivity and molar conductivity.
19. Explain why the anode is negative and the cathode is positive in a galvanic cell.
20. Find the number of coulombs required to deposit 1 mole of Al from Al_2O_3 ($\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$).
21. State why the corrosion of iron (rusting) is considered an electrochemical process.
22. Explain why increasing temperature generally increases the conductivity of an electrolytic solution.

Answer Key

1. $E(\text{cell}) = 0.34 - (-0.76) = 1.10 \text{ V}$
2. Conceptual - spontaneous, since $E(\text{cell}) > 0$
3. 2 mol electrons
4. $Q = 7200 \text{ C}$
5. Conceptual - mass deposited = ZIt , proportional to charge passed
6. $\Delta G = -nFE = -2 \times 96500 \times 1.1 = -212300 \text{ J} = -212.3 \text{ kJ}$
7. Conceptual - galvanic cell produces electrical energy from spontaneous reaction; electrolytic cell uses electrical energy to drive a non-spontaneous reaction
8. Conceptual - as reactant ion concentration decreases, Q increases, so E decreases (per Nernst equation, $E = E^\circ - (0.0591/n)\log Q$)
9. Equivalent weight = $63.5/2 = 31.75 \text{ g/equiv}$
10. Conceptual - $\Lambda_m(\infty) = \text{sum of limiting ionic conductivities}$; useful since weak electrolytes can't be extrapolated directly to infinite dilution from experimental data
11. Conceptual - SHE is assigned $E=0$ by convention, providing a universal reference point to measure other electrode potentials
12. Charge = $2 \times 96500 = 193000 \text{ C}$
13. Conceptual - dilution reduces ionic interactions, increasing ionic mobility and hence molar conductivity
14. Conceptual - negative ΔG (spontaneous) corresponds to positive $E(\text{cell})$; relation $\Delta G = -nFE$ links thermodynamics and electrochemistry
15. $E(\text{cell}) = -0.44 - (-0.76) = 0.32 \text{ V}$
16. Conceptual - salt bridge maintains electrical neutrality by allowing ion flow between half-cells, completing the circuit
17. moles of electrons = $(5 \times 7200)/96500$ approx 0.373 mol
18. Conceptual - specific conductivity is conductivity per unit volume; molar conductivity accounts for the concentration/amount of electrolyte present
19. Conceptual - anode loses electrons (oxidation, becomes electron-rich relatively... actually anode is where oxidation occurs, electrons flow out from anode to external circuit making it negative terminal)
20. Charge = $3 \times 96500 = 289500 \text{ C}$
21. Conceptual - rusting involves oxidation of Fe and reduction of O_2 , with electron flow through the metal, forming a natural electrochemical cell
22. Conceptual - higher temperature increases ionic mobility and decreases viscosity, increasing conductivity

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.